

UK Ozone

Ground-level ozone is a dangerous air pollutant that can negatively impact human health, causing respiratory irritation, coughing, and potentially worsening asthma and other respiratory conditions. We model the spatio-temporal variation of ozone across parts of England and Wales.

```
ozone <- read.csv("ozone.csv")
head(ozone)
```

```
##           site Latitude Longitude      o3 year
## 1 London Canvey 51.53308  0.567837 28.88414 1980
## 2 Central London 51.49472 -0.138333 18.35976 1980
## 3 London Harrow 51.57389 -0.352051 25.05850 1980
## 4      Sibton 52.29440  1.463497 32.88489 1980
## 5 Stevenage 51.88694 -0.200833 19.43374 1980
## 6 Bottesford 52.93028 -0.814722 37.82341 1981
```

The dataset contains the annual average ozone (o3; ppb) recorded at a named site in England and Wales, alongside the year of the recording and the Longitude and Latitude coordinates.

Bayesian regression model with a Gaussian likelihood, taking ozone as the response. Include, as fixed effects, the latitude and longitude coordinates, and the year, with SPDE random effects on both the spatial locations and the year.

```
library(INLA)
```

```
## Loading required package: Matrix
```

```
## This is INLA_24.12.11 built 2024-12-11 19:58:26 UTC.
## - See www.r-inla.org/contact-us for how to get help.
## - List available models/likelihoods/etc with inla.list.models()
## - Use inla.doc(<NAME>) to access documentation
```

```
library(inlabru)
```

```
## Warning: package 'inlabru' was built under R version 4.4.3
```

```
## Loading required package: fmesher
```

```
## Warning: package 'fmesher' was built under R version 4.4.3
```

```
library(ggplot2)
```

```
#center the year
ozone$center_year <- ozone$year - mean(ozone$year)

#spatial coordinates
locations <- data.frame(easting=ozone$Longitude, northing=ozone$Latitude )

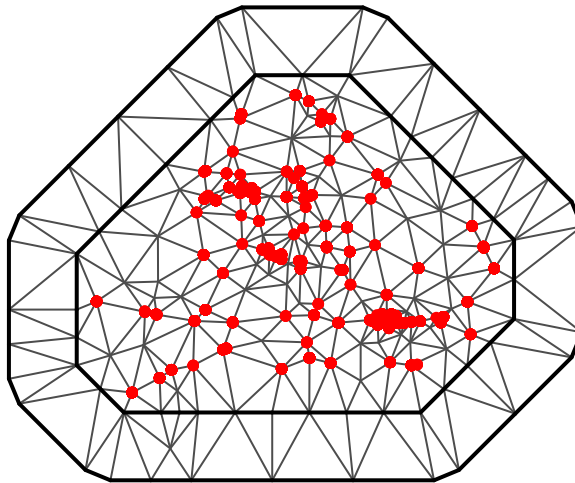
#spatial mesh
loc_mesh <- inla.mesh.2d(
  locations,
```

```

max.edge = c(1, 2),
cutoff = 0.1      #to avoid duplicated points and not points that are too close
)

#plot mesh
plot(loc_mesh)
points(locations, col = "red", pch = 20)

```



```

#location SPDE with PC priors
loc_spde <- inla.spde2.pcmatern(
  mesh = loc_mesh,
  prior.range = c(1, 0.5), #P(range < 1) = 0.5
  prior.sigma = c(1, 0.5) #P(sigma > 1) = 0.5
)

#centered year: 1D mesh
mesh_year <- inla.mesh.1d(
  seq(min(ozone$center_year), max(ozone$center_year), length.out = 100))

#year SPDE with PC priors
spde_time <- inla.spde2.pcmatern(
  mesh = mesh_year,
  prior.range = c(1, 0.5),
  prior.sigma = c(1, 0.5))

```

```
ozone$geometry <- sf::st_as_sf(locations, coords = c("easting", "northing"))$geometry
```

```
cmp <- o3 ~ Intercept(1)+ Longitude + Latitude + center_year +  
      floc(geometry, model = loc_spde) + ft(center_year, model = spde_time)
```

```
#fit model
```

```
q2_m1_bru <- bru(components = cmp, family = "gaussian", data = ozone,  
                formula = ~ Intercept + Longitude + Latitude + center_year  
                + floc + ft, options=list(verbose = T))
```

```
summary(q2_m1_bru)
```

```
## inlabru version: 2.12.0
## INLA version: 24.12.11
## Components:
## Intercept: main = linear(1), group = exchangeable(1L), replicate = iid(1L), NULL
## Longitude: main = linear(Longitude), group = exchangeable(1L), replicate = iid(1L), NULL
## Latitude: main = linear(Latitude), group = exchangeable(1L), replicate = iid(1L), NULL
## center_year: main = linear(center_year), group = exchangeable(1L), replicate = iid(1L), NULL
## floc: main = spde(geometry), group = exchangeable(1L), replicate = iid(1L), NULL
## ft: main = spde(center_year), group = exchangeable(1L), replicate = iid(1L), NULL
## Likelihoods:
##   Family: 'gaussian'
##   Tag: ''
##   Data class: 'data.frame'
##   Response class: 'numeric'
##   Predictor:
##     o3 ~ Intercept + Longitude + Latitude + center_year + floc +
##         ft
##   Used components: effects[Intercept, Longitude, Latitude, center_year, floc, ft], latent[]
## Time used:
##   Pre = 3.23, Running = 0.956, Post = 0.901, Total = 5.09
## Fixed effects:
##           mean      sd 0.025quant 0.5quant 0.975quant   mode kld
## Intercept   2.351 26.992   -50.776    2.414    55.126   2.415  0
## Longitude  -1.210  0.826    -2.846   -1.207     0.413  -1.207  0
## Latitude     0.836  0.515    -0.172    0.835     1.851   0.835  0
## center_year 0.334  0.061     0.212    0.335     0.454   0.335  0
##
## Random effects:
##   Name      Model
##   floc SPDE2 model
##   ft  SPDE2 model
##
## Model hyperparameters:
##           mean      sd 0.025quant 0.5quant
## Precision for the Gaussian observations 0.049 0.002    0.046  0.049
## Range for floc      0.468 0.074    0.341  0.462
## Stdev for floc      9.271 0.726    7.934  9.240
## Range for ft        2.072 0.484    1.294  2.013
## Stdev for ft        3.209 0.475    2.395  3.168
##           0.975quant   mode
```

```

## Precision for the Gaussian observations      0.052 0.049
## Range for floc                             0.632 0.449
## Stdev for floc                            10.792 9.169
## Range for ft                               3.190 1.896
## Stdev for ft                              4.261 3.076
##
## Deviance Information Criterion (DIC) .....: 13192.76
## Deviance Information Criterion (DIC, saturated) .....: 2363.96
## Effective number of parameters .....: 131.16
##
## Watanabe-Akaike information criterion (WAIC) ....: 13199.48
## Effective number of parameters .....: 127.85
##
## Marginal log-Likelihood: -6813.74
## is computed
## Posterior summaries for the linear predictor and the fitted values are computed
## (Posterior marginals needs also 'control.compute=list(return.marginals.predictor=TRUE)')

```

To investigate the spatial and temporal variation in ozone concentration across England and Wales, we fitted a Bayesian regression model using `inlabru`. The model assumes a Gaussian likelihood and uses both fixed and SPDE random effects.

The fixed effects keep track linear trends in longitude, latitude, and year. The year variable has been centred to improve numerical stability.

Insights from the fixed effects:

- Intercept: The posterior mean of the intercept is approximately 2.30, but the 95% credible interval is extremely wide (approximately -50.9 to 55.1).
This is expected, as the intercept corresponds to the predicted ozone level when all covariates are set to zero, in our case that is at longitude = 0, latitude = 0, and the mean year. However, the geographic coordinates (0,0) lie far outside the domain of the data (which covers England and Wales). The point (0,0) is located in the Atlantic Ocean near the Gulf of Guinea. Since the model must work on an unobserved region to define the intercept, the estimate shows high uncertainty. In this case, the intercept should not be interpreted alone.
- Longitude has a posterior mean around -1.21 , with a 95% credible interval that goes from -2.85 to 0.42 . This negative association suggests that, on average, ozone concentrations tend to decrease by about 1.2 ppb for each unit increase in longitude (moving east). Since the credible interval contains 0 we cannot be too sure about this.
- Latitude has a posterior mean of around 0.84, with a 95% credible interval from -0.17 to 1.85 . This indicates a positive but, on average, weaker effect, compared to longitude. This suggests that ozone levels increases moving north, but again the credible interval includes zero, so this trend is not something to be too sure about.
- Centered year shows a posterior mean of 0.33 (95% CI: 0.21 to 0.45), meaning that ozone levels have increased over time. This corresponds to an increase of approximately 0.33 ppb per year. The credible interval does not include zero, indicating that this trend is statistically significant.

To model spatial and temporal effects, we introduced two random effects using the SPDE approach:

- A spatial SPDE defined over a 2D mesh built on longitude and latitude. This allows to capture local behaviour in ozone levels across space.
- A temporal SPDE defined over a 1D mesh built on the centered year. This allows to capture non-linear trends over time.

Both SPDE components use PC priors of the form $PC(1, 0.5)$, meaning the prior places a 50% probability that the range is smaller than 1 (for both spatial and temporal fields) and a 50% probability that the standard deviation exceeds 1.

To build the spatial mesh, we tested several values of the cutoff argument. Small values tend to build a mesh that gives back really localised effects, due to high point concentration in some areas. Setting cutoff to 0.1 provides a reasonable trade-off: it removes points closer than 0.1 units (roughly 10 km), which is appropriate given the geographic scale of the study area (England and Wales).

The spatial field explains geographic variability that is not explained by the fixed effects: the posterior mean for the spatial range is around 0.47, with a 95% credible interval that goes from around 0.34 to 0.64. This suggests that spatial correlation is gone after around 0.5 units in the coordinate space. The quite narrow credible interval shows that the model is confident about the scale of spatial dependence.

Although the posterior mean for the spatial range is around 0.47, this value should be interpreted in the context of the coordinates in the dataset. In our data, both latitude and longitude span a range of around 5 units, so a spatial range of 0.47 represents roughly 10% of each spatial dimension.

The posterior standard deviation of the spatial field has a mean of 9.26, with a 95% credible interval from around 7.9 to 10.7. This implies a quite high degree of variability, considering the range of ozone in our dataset is from 9.83 to 87.1.

Regarding the random effect over the years:

The posterior mean for the temporal range is approximately 2.0, with a 95% credible interval from around 1.2 to 3.3, indicating that temporal correlation is present over a period of around 1.5 to 3 years.

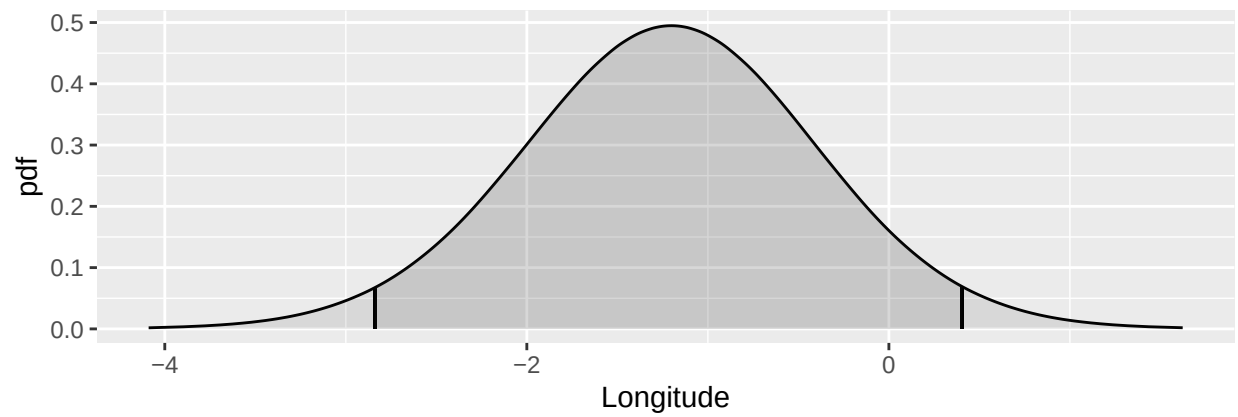
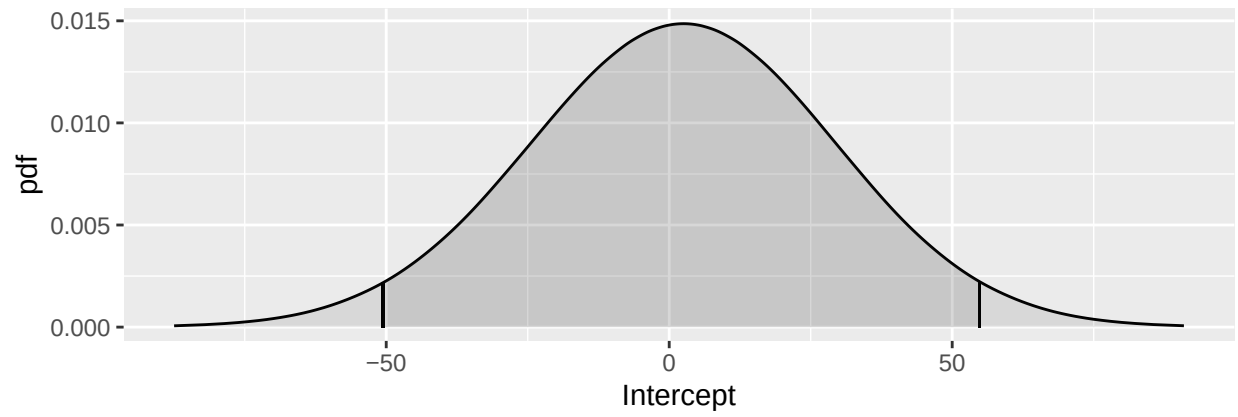
The posterior mean for the standard deviation is 3.20 (95% CI: 2.4 to 4.2), which reflects a moderate degree of temporal variability in ozone levels. While this variability is meaningful, it is not as strong as the spatial one, meaning that spatial differences across locations contribute more strongly to variation in ozone than temporal changes.

```
int_plot <- plot(q2_m1_bru, "Intercept")
long_plot <- plot(q2_m1_bru, "Longitude")
lat_plot <- plot(q2_m1_bru, "Latitude")
year_plot <- plot(q2_m1_bru, "center_year")

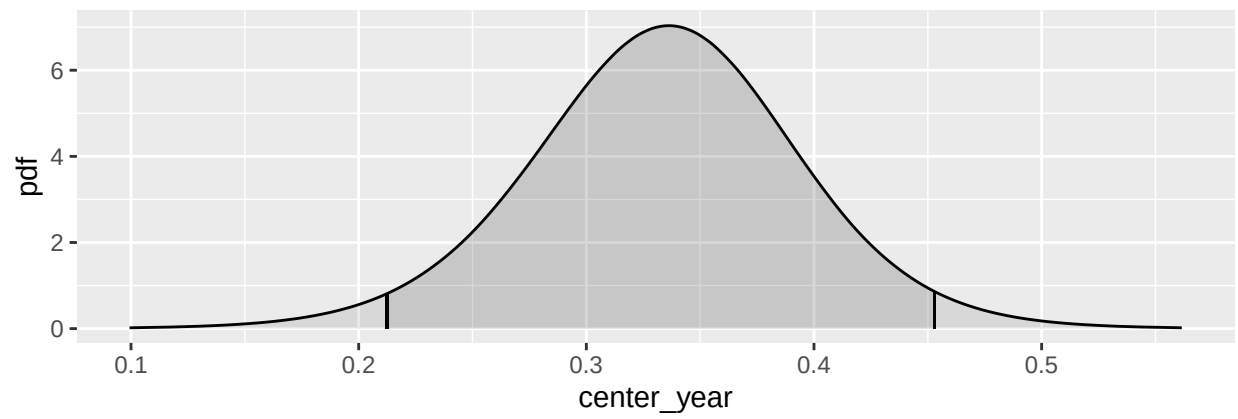
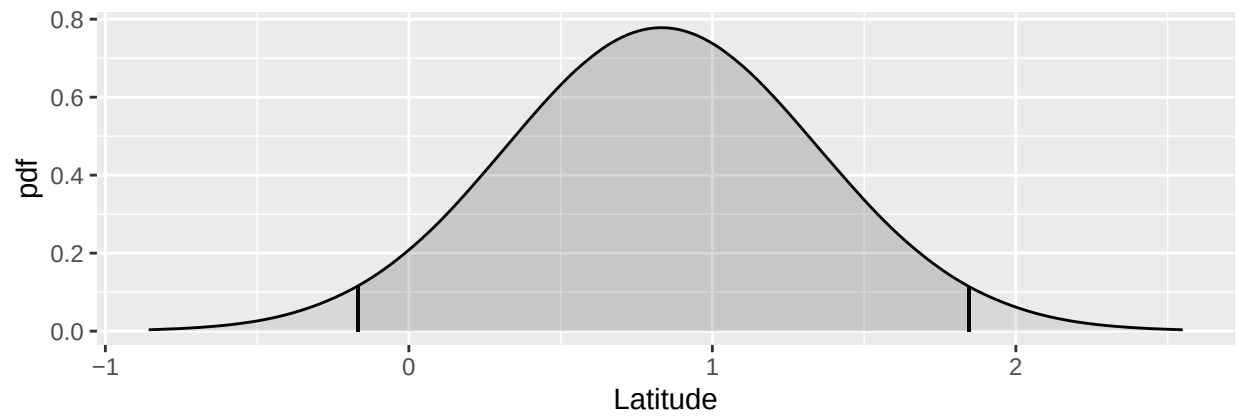
spde_ft_range <- spde.posterior(q2_m1_bru, "ft", what = "range")
spde_ft_logvar <- spde.posterior(q2_m1_bru, "ft", what = "log.variance")

range_ft_plot <- plot(spde_ft_range)
var_ft_plot <- plot(spde_ft_logvar)

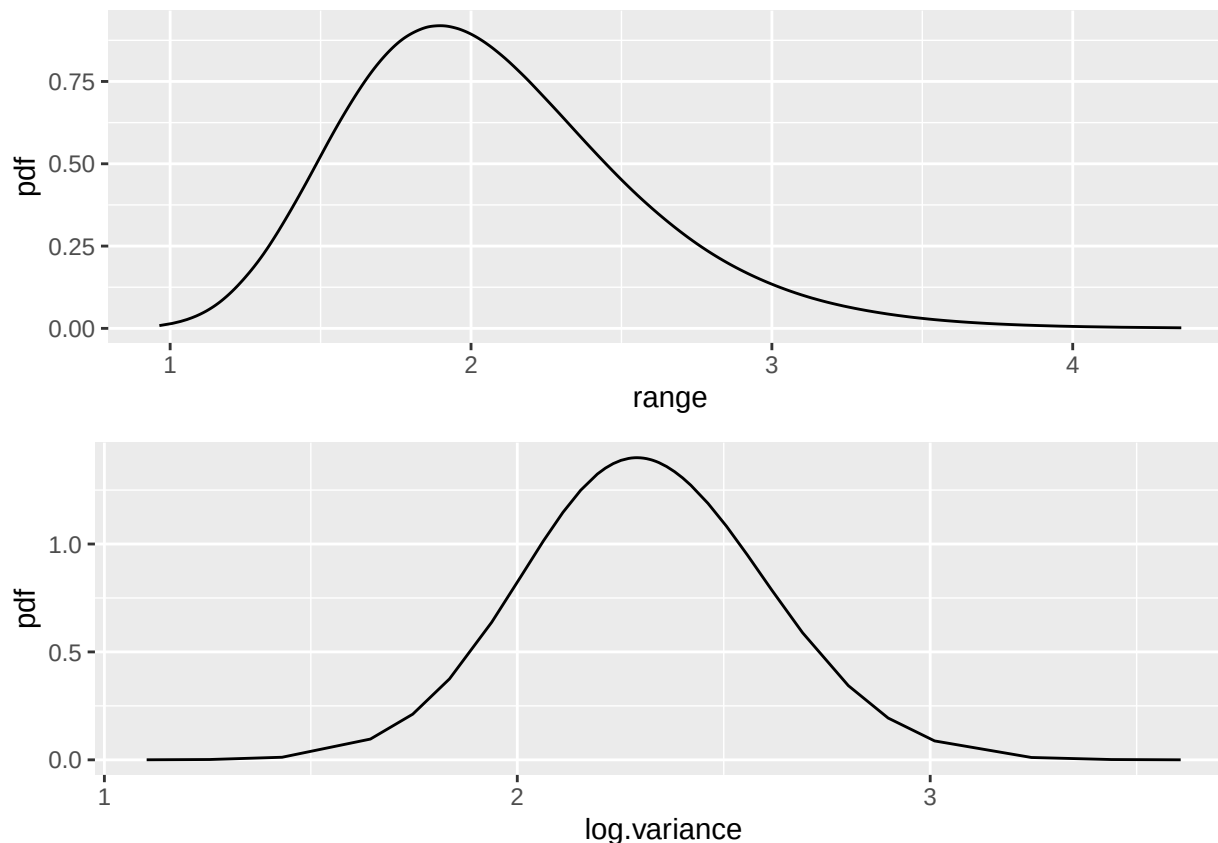
multiplot(int_plot, long_plot)
```



```
multiplot(lat_plot, year_plot)
```



```
multiplot(range_ft_plot, var_ft_plot)
```



The marginal posterior distributions for all fixed effects and the parameters of the temporal SPDE effect are shown above.

The intercept shows a very wide distribution, confirming that it is weakly identified due to the location (0,0) outside the observed data range that we talked about before.

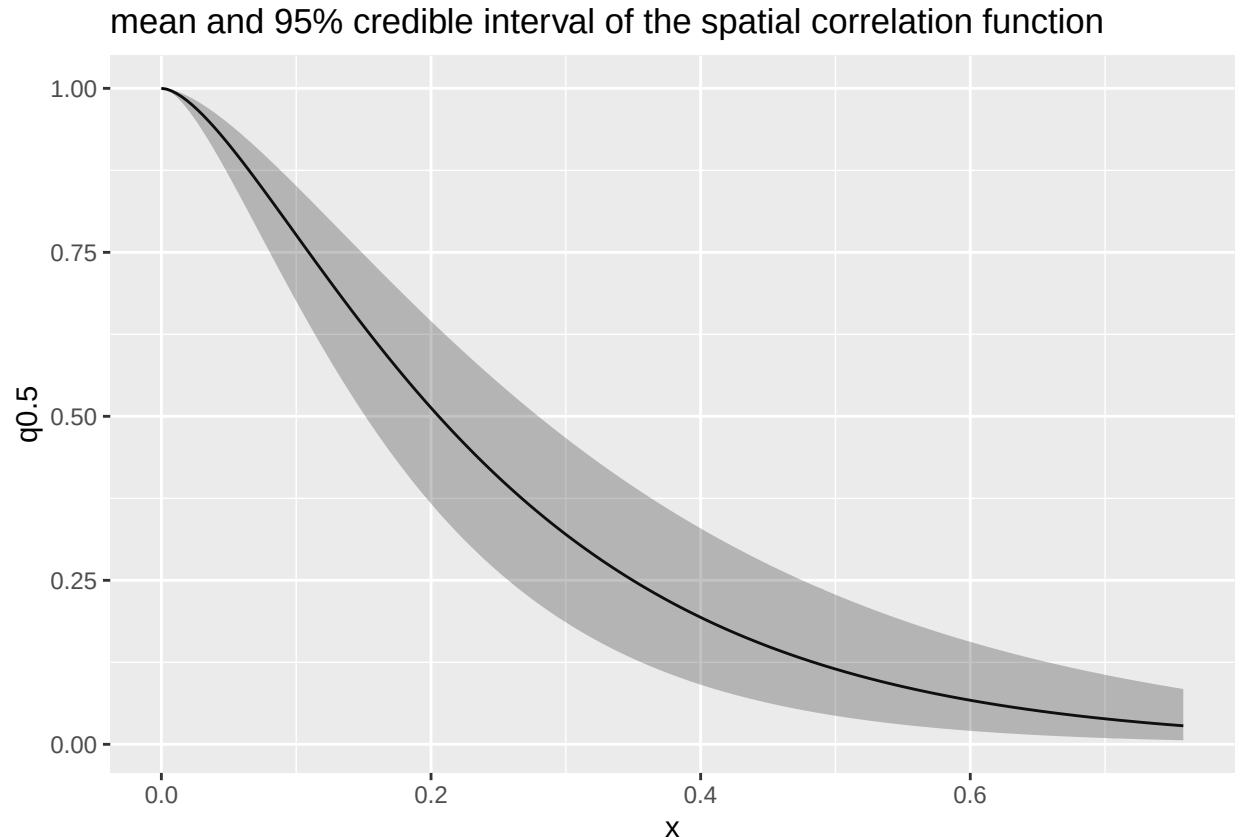
The distributions for longitude and latitude are more concentrated, with longitude showing with most probability a negative effect and latitude showing a weak positive one. Again, both of the domain of the posterior distribution (and their 95% CIs contain 0).

The posterior for center_year has positive domain, indicating a statistically significant increasing trend in ozone concentration over the years.

The temporal SPDE parameters:

The range distribution is right-skewed, with most of the probability in the range 1.5 to 3 years, indicating that the temporal correlation extends across approximately this time period, while the log.variance shows a symmetric posterior, with most of the probability in the range 2-3. The shape and domain of these distributions suggest that the model is learning the temporal structure from the data with good confidence.

```
corplot_space <- spde.posterior(q2_m1_bru, "floc", what = "matern.correlation")
plot(corplot_space)+
ggtitle("mean and 95% credible interval of the spatial correlation function")
```



The plot above shows the estimated posterior mean (black line) and 95% credible intervals (grey area) of the spatial correlation function from the model.

As expected for a Matérn correlation, the function decreases from 1 toward 0 as distance increases.

The correlation drops below 0.5 within approximately 0.2 units, and below 0.1 within 0.5 units, confirming the relatively short range that we talked about previously. The credible interval is narrower for small distances (x axis < 0.1), where the data are more informative.

To assess the robustness of this estimate (next block of code), we tested the sensitivity of the spatial correlation function to different modelling choices:

- The prior choice on the SPDE range and standard deviation (for the spatial effect)
- The resolution of the spatial mesh

```
#let's write a function to run the model using different priors and resolution

#looking at the mesh it seemed already quite narrow,
#let's try to make it wider (less resolution)
mesh_big <- inla.mesh.2d(locations, max.edge = c(2, 4), cutoff = 0.1 )
mesh_std  <- inla.mesh.2d(locations, max.edge = c(1, 2), cutoff = 0.1)

#different priors
prior_std <- list(range = c(1, 0.5), sigma = c(1, 0.5)) #original priors
prior_alt <- list(range = c(2, 0.5), sigma = c(4, 0.5)) #new priors

#function to fit the model
```

```

fit_model <- function(mesh, prior) {

  spde <- inla.spde2.pcmatern(
    mesh = mesh,
    prior.range = prior$range,
    prior.sigma = prior$sigma)

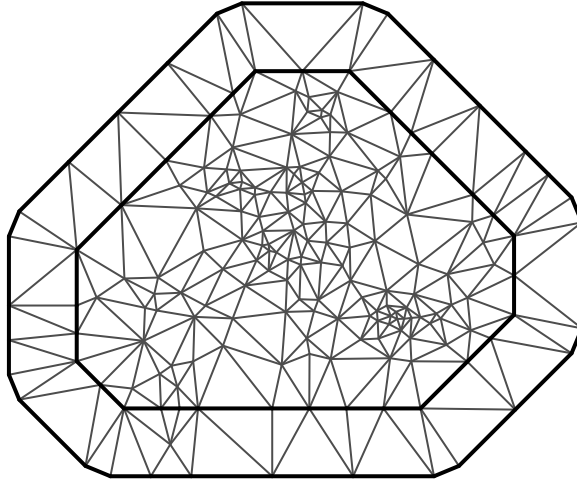
  #always same temporal mesh
  mesh_time <- inla.mesh.1d(
    seq(min(ozone$center_year), max(ozone$center_year), length.out = 100))
  spde_time <- inla.spde2.pcmatern(
    mesh = mesh_time,
    prior.range = c(1, 0.5),
    prior.sigma = c(1, 0.5))

  cmp <- o3 ~ Intercept(1) + Longitude + Latitude + center_year +
    floc(geometry, model = spde) + ft(center_year, model = spde_time)

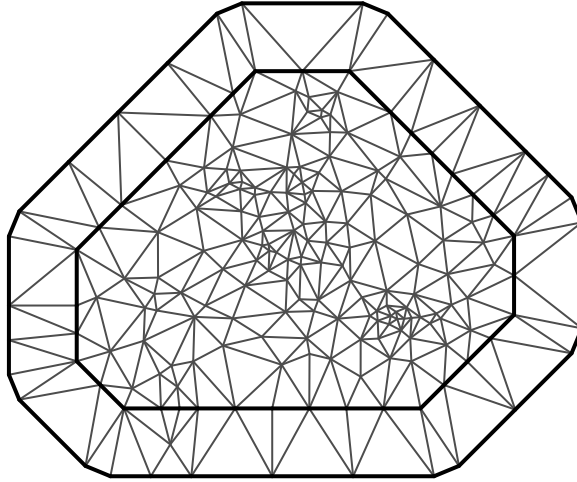
  bru(
    cmp,
    family = "gaussian",
    data = ozone,
    formula = ~ Intercept + Longitude + Latitude + center_year + floc + ft,
    options = list(verbose = TRUE)
  )
}

#compared mesh
plot(mesh_big) #new

```



```
plot(loc_mesh) #original
```



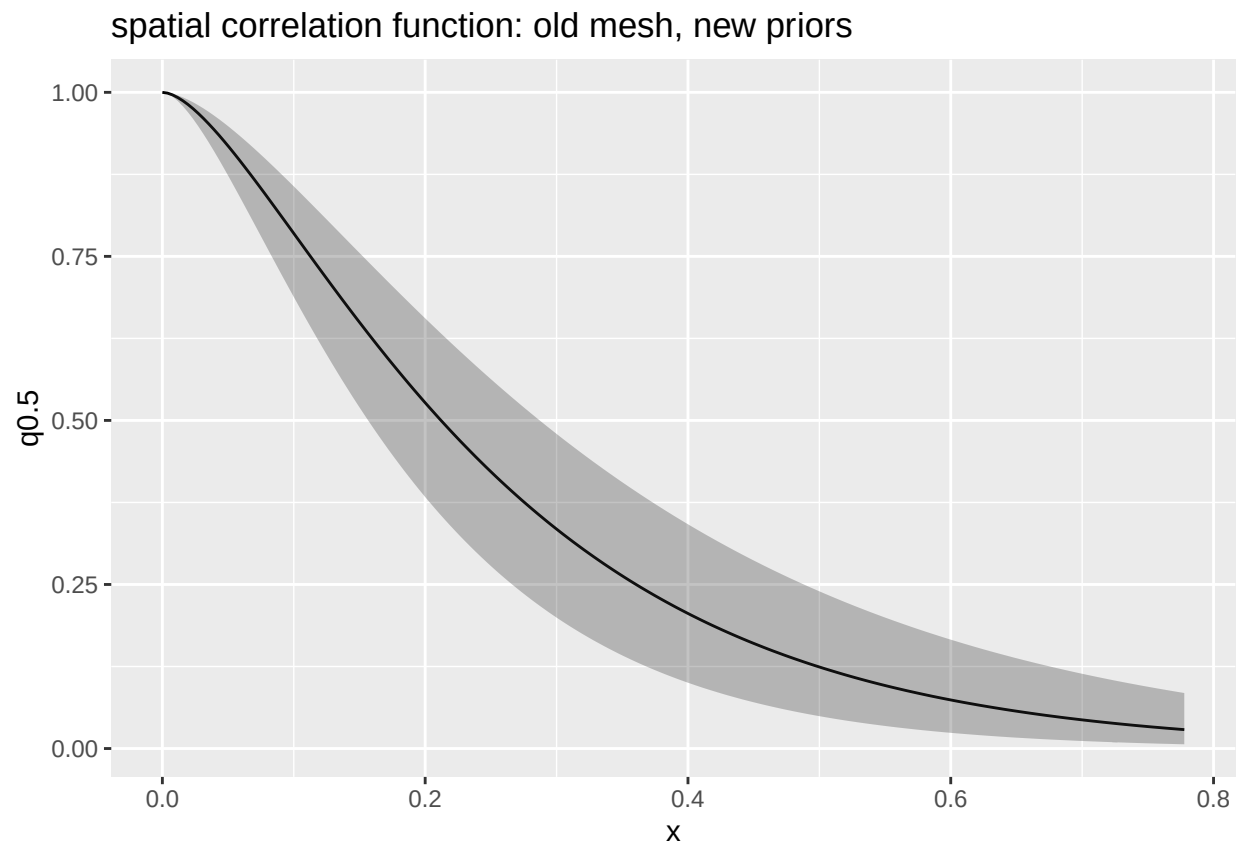
```
#fitting
mod_std_big <- fit_model(mesh_big, prior_std) #mesh big & std priors
mod_std_alt  <- fit_model(mesh_std, prior_alt) #mesh std & new priors
mod_new <- fit_model(mesh_big, prior_alt) #mesh big & new priors

cor_big_mesh <- spde.posterior(mod_std_big, "floc", what = "matern.correlation")

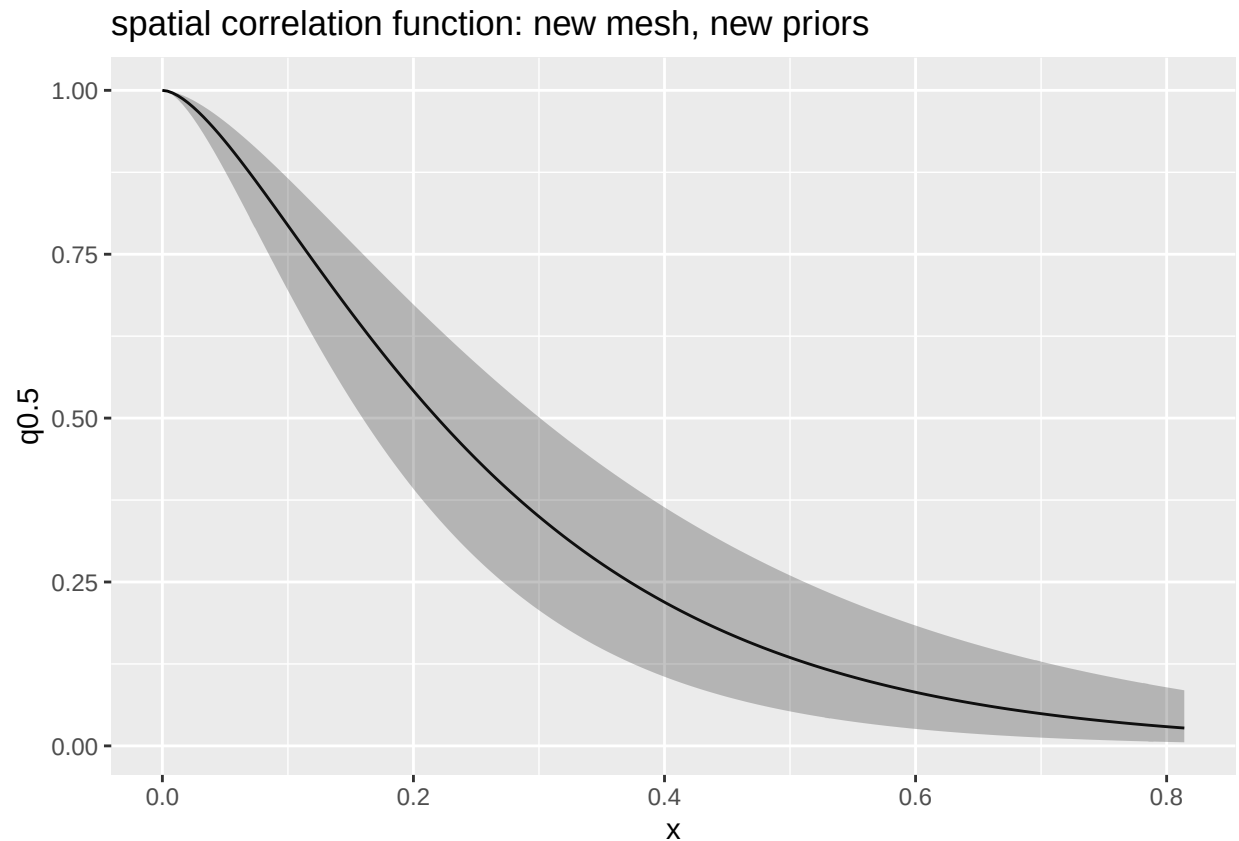
plot(cor_big_mesh)+
  ggtitle("spatial correlation function: new mesh, old priors")
```



```
cor_new_prior <- spde.posterior(mod_std_alt, "floc", what = "matern.correlation")  
plot(cor_new_prior)+  
  ggtitle("spatial correlation function: old mesh, new priors")
```



```
cor_all_new <- spde.posterior(mod_new, "floc", what = "matern.correlation")  
plot(cor_all_new)+  
  ggtitle("spatial correlation function: new mesh, new priors")
```



```
#all together:

#names for labels
corplot_space$model <- "original mesh & priors"

cor_big_mesh$model <- "new mesh, old priors"

cor_new_prior$model <- "old mesh, new priors"

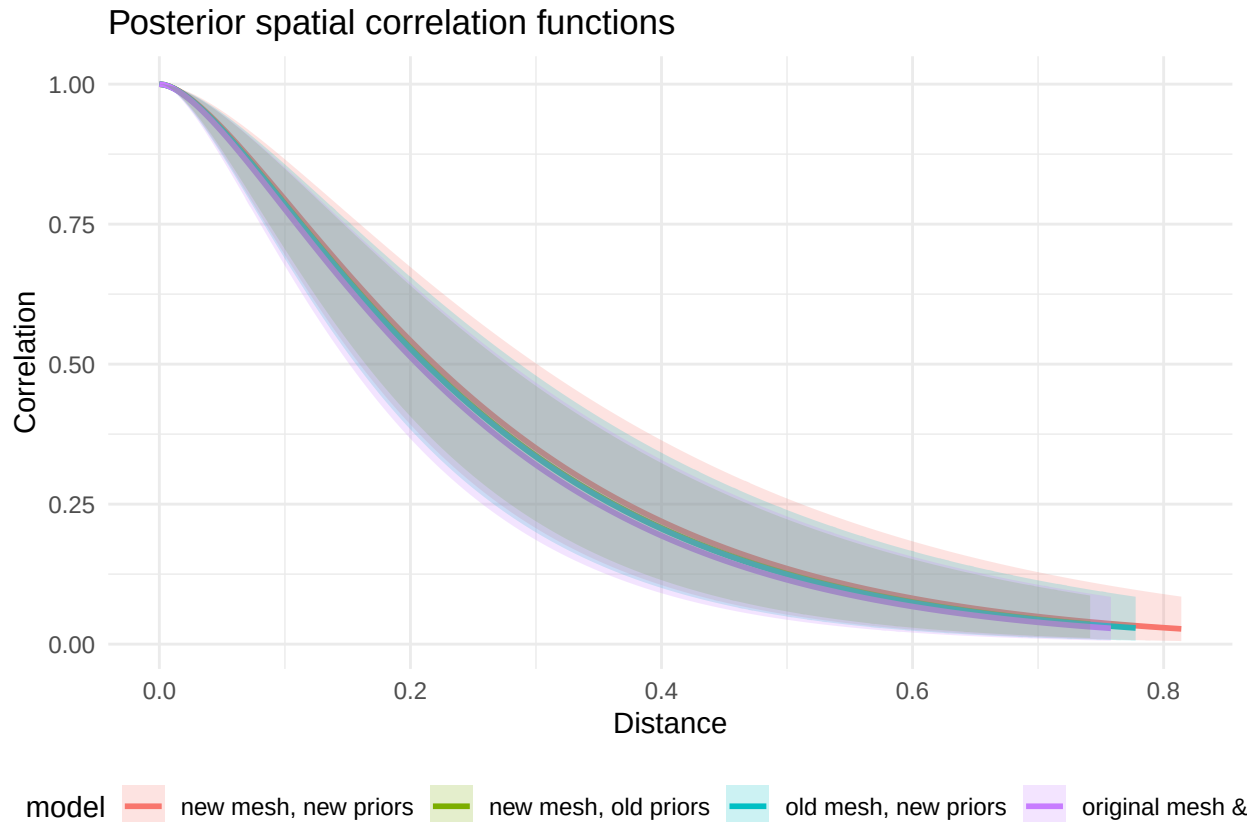
cor_all_new$model <- "new mesh, new priors"

#dataframe for all
corr_df_all <- rbind(corplot_space, cor_big_mesh, cor_new_prior, cor_all_new)

#plot together
ggplot(corr_df_all, aes(x = x, y = q0.5, color = model, fill = model)) +
  geom_line(size = 1) +
  geom_ribbon(aes(ymin = q0.025, ymax = q0.975), alpha = 0.2, color = NA) +
  labs(
    title = "Posterior spatial correlation functions",
    x = "Distance",
    y = "Correlation"
  ) +
  theme_minimal() + theme(legend.position = "bottom")
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
```

```
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



To evaluate the robustness of the estimated spatial correlation function, we focused on two components:

1. The PC priors for the SPDE spatial parameters.

The original model used $PC(1, 0.5)$ priors, corresponding to $P(\text{range} < 1) = 0.5$ and $P(\text{sd} > 1) = 0.5$. Given that the posterior mean for the spatial range was relatively low (around 0.47), we tested a less restrictive prior with $P(\text{range} < 2) = 0.5$. Similarly, since the posterior standard deviation was high (around 9.27), we tested a more restrictive prior with $P(\text{sd} > 4) = 0.5$, reducing probability of large variability.

2. The resolution of the spatial mesh.

The original mesh ($\text{max.edge} = c(1, 2)$) was relatively “small”, adapted to the data locations. We tested a “wider” mesh ($\text{max.edge} = c(2, 4)$) to see if reducing the resolution would affect the correlation.

Despite the changes, the overall shape and behaviour of the correlation function remain stable.

The original estimate of the correlation (purple) lies very close to the other curves, suggesting that the spatial dependence structure is driven by the data rather than the modelling assumptions.

```
#years (go back to original scale of the years)
years_or <- mesh_year$loc + mean(ozone$year)
```

```

# fixed
fixed_effect <- q2_m1_bru$summary.fixed$mean[4] * mesh_year$loc

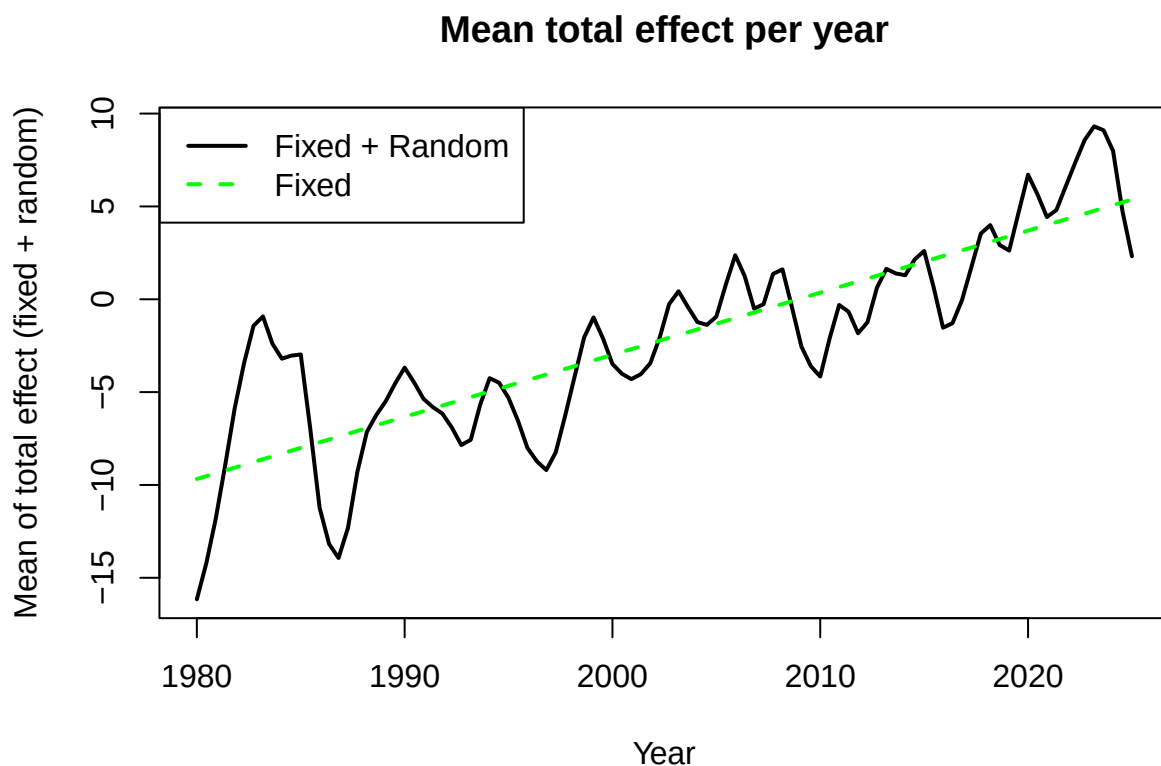
#fixed + random
total_effect <- fixed_effect + q2_m1_bru$summary.random$ft$mean

#plot total effect
plot(years_or, total_effect,
     type = "l", col = "black", lwd = 2,
     xlab = "Year", ylab = "Mean of total effect (fixed + random)",
     main = "Mean total effect per year")

#lines for only fixed effect
lines(years_or, fixed_effect, col = "green", lwd = 2, lty = 2)

legend("topleft", legend = c("Fixed + Random", "Fixed"),
      col = c("black", "green"), lty = c(1, 2), lwd = 2)

```



The plot shows the posterior mean of the **total** temporal effect (black line) and the posterior mean of the temporal **fixed** effect (green line).

The total effect (SPDE random + fixed) displays a clear increasing trend, with fluctuations, from the late 1980s onward, suggesting an increase in ozone levels over time.

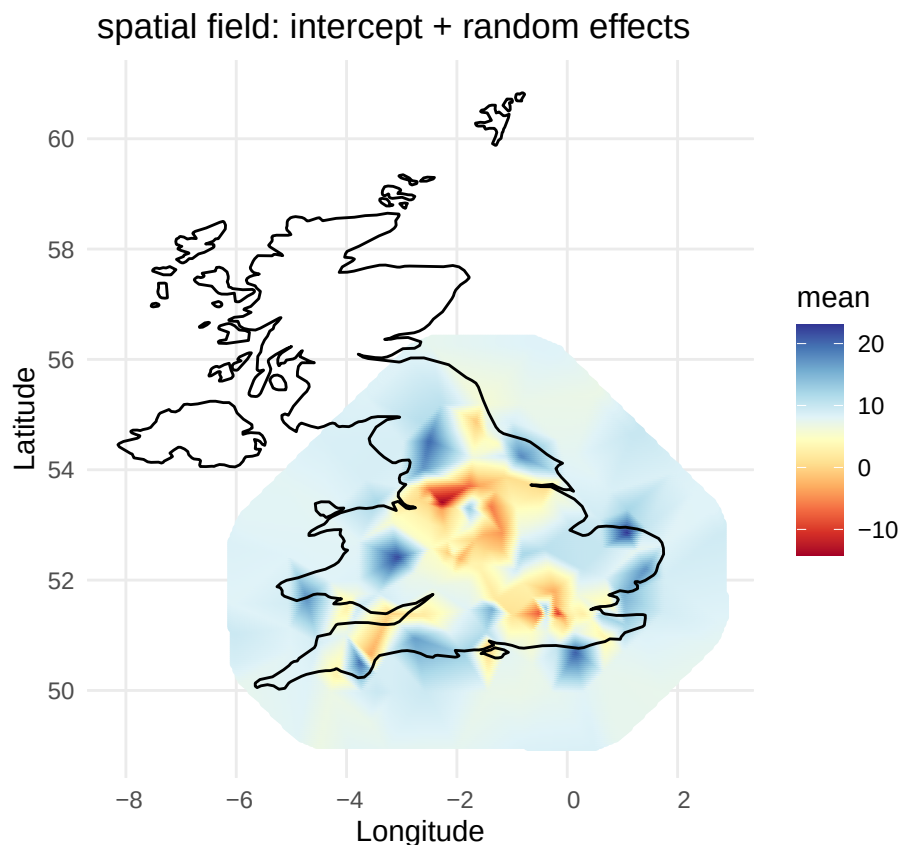
Since the year covariate is centered (so baseline is the mean year, that is around 2009), we can see from the plot that early years (1980s and early 1990s) show negative total effects, indicating negative impact compared to 2009. In contrast, more recent years (2010s and early 2020s) show positive values.

This behaviour could be explained by factors, such as pollution, changes in emissions or in atmospheric conditions. The presence of the nonlinear random effect allows the model to explain fluctuations that the fixed linear trend alone would not explain.

```
library(RColorBrewer)
UK <- map_data(map = "world", region = "UK") # changed map to "world"
#spatial effect (intercept + floc)
pix <- fm_pixels(loc_mesh, dims = c(200, 200),format = "sf")

pred <- predict(q2_m1_bru, pix, formula=~floc+Intercept)

ggplot() +
  gg(loc_mesh) +
  gg(pred, aes(color = mean))+
  scale_color_gradientn(colours=brewer.pal(11, "RdYlBu"),limits=range(pred$mean)) +
  geom_polygon(data = UK, aes(x = long, y = lat, group = group), fill = NA, color = "black")+
  labs(
    title = " spatial field: intercept + random effects",
    x = "Longitude", y = "Latitude"
  ) +
  theme_minimal()
```



The map shows the posterior mean of the spatial effect obtained by combining the intercept and the spatial random effect .

The plot shows spatial variability with alternating zones of positive and negative deviations from the average. In the following section, we analyse the full predicted field including both fixed and random effects.

```

#spatial effect ()

stpix = sf::st_coordinates(pix)
pix$Latitude = stpix[,2]
pix$Longitude = stpix[,1]

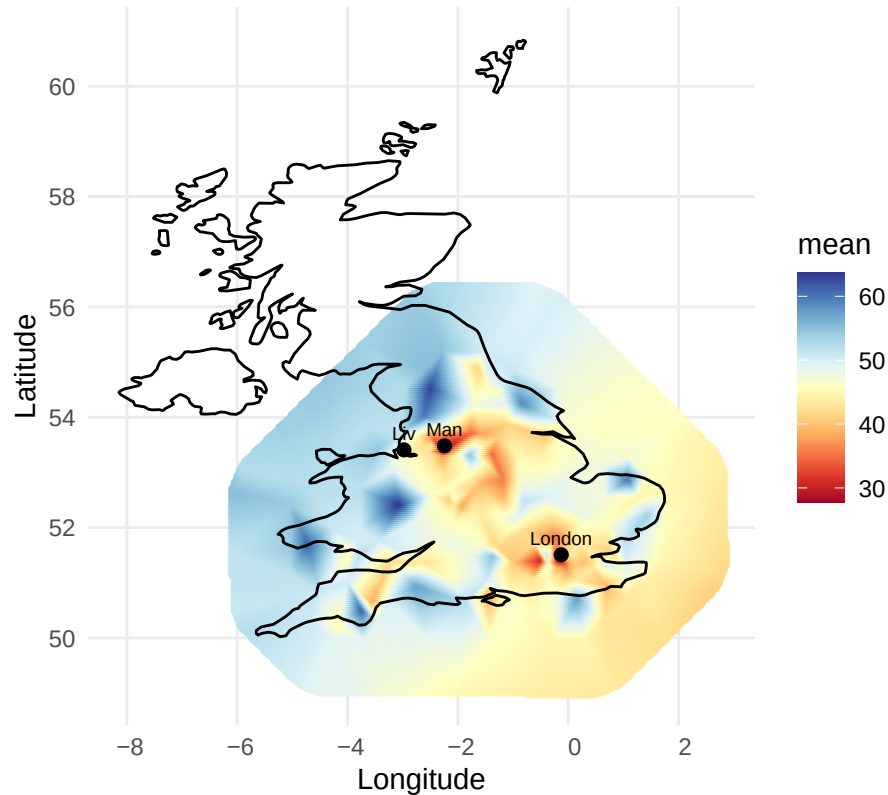
#predict full spatial effect: Intercept + Longitude + Latitude + floc
pred_total <- predict(q2_m1_bru, pix, formula = ~ Intercept + Longitude +
                      Latitude + floc)

# coordinates big and industrials city UK
cities <- data.frame(
  name = c("London", "Man", "Liv"),
  x = c(-0.1276, -2.2426, -2.9785),
  y = c(51.5072, 53.4808, 53.4084))

#plot
ggplot() +
  gg(loc_mesh) +
  gg(pred_total, aes(color = mean)) +
  scale_color_gradientn(
    colours = brewer.pal(11, "RdYlBu"),
    limits = range(pred_total$mean)) +
  geom_polygon(data = UK, aes(x = long, y = lat, group = group),
    fill = NA, color = "black") +
  geom_point(data = cities, aes(x = x, y = y), color = 'black', size = 2) +
  geom_text(data = cities, aes(x = x, y = y, label = name),
    color = 'black', size = 2.5, nudge_y = 0.3) +
  labs(
    title = "spatial field: fixed + random effects",
    x = "Longitude", y = "Latitude"
  ) +
  theme_minimal()

```

spatial field: fixed + random effects



The map displays the posterior mean of the full spatial prediction for ozone levels, combining both fixed effects (longitude and latitude) and the spatial random field (floc), along with the intercept.

The inclusion of fixed effects introduces a visible east-west difference, east shows lower level of ozone, as we have seen before from the summary.

Known big and industrial cities of the UK are plotted as well, but interestingly they do not show high values of ozone, let's try and see why:

The spatial and temporal patterns estimated by our model align with current knowledge about ozone dynamics in the UK. (source: <https://www.gov.uk/government/statistics/air-quality-statistics/concentrations-of-ozone>).

While one might expect high ozone levels over big cities, the model reveals that these locations often do not show these high values. This is consistent with the known behaviour of ground-level ozone: urban emissions of nitrogen oxides (NO_x), can block ozone formation through chemical reactions. As a result, cities may exhibit lower ozone concentrations, while nearby suburban and rural areas, downwind of cities, often experience higher levels due to transported ozone.

The temporal effect captured by our model also reflects real-world trends. According to UK air quality reports, ozone concentrations in urban areas have risen steadily since the late 2010s, reaching record levels in 2023 (similar peak can be seen in the previous plot of our model). This increase is attributed to multiple factors, including reductions in NO_x emissions, which reduce ozone destruction, and climatic conditions such as higher temperatures and more sunlight, both of which increase ozone formation.

In summary, our model's outputs are consistent with known behaviour of ozone in the UK context.